Aim

To synthesize a discrete-time periodic square wave using discrete-time fourier series in MATLAB.

Theory

Discrete-Time Fourier Series (DTFS): A discrete-time periodic signal x[n] with fundamental period N can be represented as a DTFS. Unlike the continuous-time case, the DTFS representation is an exact finite summation & does not exhibit Gibbs phenomenon.

The analysis equation is:

$$a_k = \frac{1}{N} \sum_{n=0}^{N-1} x[n] e^{\frac{-j2\pi kn}{N}}, k = 0, 1, 2, \ldots, N-1$$

The synthesis equation is:

$$x[n] = \sum_{k=0}^{N-1} a_k e^{j\frac{2\pi}{n}kn}$$

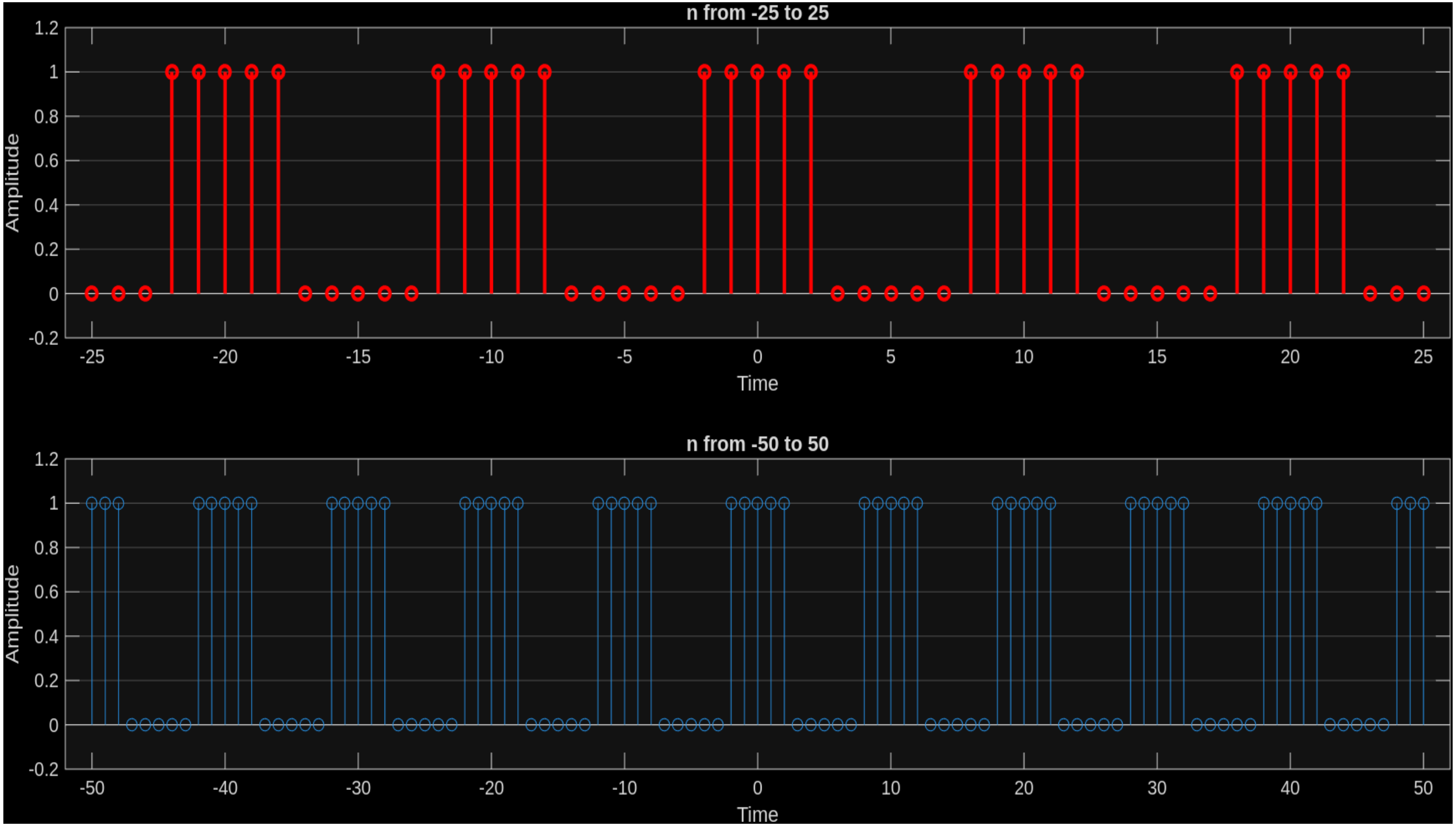
The square wave is defined as:

$$x[n] = \{ \begin{matrix} 1, |n| \le N_1 0 \\ 0, N_1 < |n| \le [\frac{N}{2}] \end{matrix}$$

MATLAB Code

```
clc
clear all
close all

N = 10;
n = 0:N-1;

k = 0:N-1;

C = (2/N) * (cos (4 * pi * (k/N))) + (2/N) * (cos (2 * pi * (k/N))) + 1/N;

    % E = exp( ( (-sqrt(-1) * 2 * pi) / N) * (0: N-1) .* (0: N-1) );

% -------------------------

n_changed_1 = (-25:25);

x = zeros(1, length(n_changed_1));


for m = 1:N
    x = x + C(m) * exp(1i * 2 * pi * n_changed_1 * (m-1) / N);
end



% -------------------------

n_changed_2 = (-50:50);

y = zeros(1, length(n_changed_2));


for m = 1:N
    y = y + C(m) * exp(1i * 2 * pi * n_changed_2 * (m-1) / N);
end

% ------------------------   m

figure;

title('DT periodic square wave using DTFS')

subplot(2, 1, 1)
stem(n_changed_1, x, LineWidth=2, Color='r')
title('n from -25 to 25')
ylabel('Amplitude')
xlabel('Time')
set(gca, 'YGrid', 'on', 'XGrid', 'off')
% grid minor;

subplot(2,1,2)
stem(n_changed_2, y)
title('n from -50 to 50')
ylabel('Amplitude')
xlabel('Time')
% grid minor;
set(gca, 'YGrid', 'on', 'XGrid', 'off')
```

Results



n from -25 to 25
1.2
1
0.8
0.6
0.4
0.2
0
-0.2
Amplitude
-25
-20
-15
-10
-5
0
5
10
15
20
25
Time
n from -50 to 50
1.2
1
0.8
0.6
0.4
0.2
0
-0.2
Amplitude
-50
-40
-30
-20
-10
0
10
20
30
40
50
Time

Conclusion

We observe that a square wave can be obtained from approximation from a DTFS to obtain a DT square wave function. We also note that we do not see any over-shoot and under-shoot phenomenon (Gibbs phenomenon).